

AGARD, Hungary

WMO: 128460

Lat: 47.183N

Lon: 18.617E

Elev: 106

StdP: 100.06

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.3	-8.3	-14.8	1.0	-10.0	-12.2	1.3	-6.9	12.4	3.2	11.0	3.2	1.7	320	0.474

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	32.9	21.8	31.2	21.1	29.6	20.5	22.8	30.1	22.0	29.2	21.2	28.2	2.4	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.4	15.3	26.1	19.5	14.5	25.3	18.7	13.7	24.6	67.7	30.2	64.5	29.2	61.8	28.4	25.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.6	9.0	7.5	-14.9	35.3	4.1	1.5	-17.8	36.4	-20.2	37.3	-22.5	38.2	-25.4	39.3		
(5)			-15.2	24.3	4.0	1.0	-18.1	25.0	-20.4	25.6	-22.7	26.2	-25.6	26.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.4	-0.2	1.5	6.4	12.3	16.5	20.5	22.1	21.7	16.8	11.6	6.3	1.0		
	DBStd	8.77	4.29	4.61	4.31	3.65	3.43	3.40	3.06	3.14	3.43	3.89	3.84	4.28		
	HDD10.0	1127	316	238	123	18	1	0	0	0	29	120	281			
	HDD18.3	2878	574	471	369	182	76	17	4	7	70	209	360	539		
	CDD10.0	1648	0	0	13	88	203	314	376	362	203	79	10	1		
	CDD18.3	359	0	0	0	2	20	81	122	111	23	1	0	0		
	CDH23.3	3506	0	0	0	36	215	778	1171	1077	216	12	0	0		
	CDH26.7	1236	0	0	0	5	40	256	453	428	53	0	0	0		
(14) Wind	WSAvg	2.8	2.7	3.0	3.3	3.1	3.1	2.8	2.8	2.5	2.5	2.3	2.5	2.5		
(15) Precipitation	PrecAvg	552	33	29	30	38	52	67	59	59	52	42	47	44		
	PrecMax	918	71	69	92	93	166	174	127	201	141	97	104	99		
	PrecMin	278	2	0	1	2	11	6	7	6	4	0	0	0		
	PrecStd	146	20	22	21	23	32	43	33	45	38	29	24	24		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.4	14.5	20.7	27.3	29.8	33.2	35.1	35.4	30.7	25.0	18.6	12.9		
		MCWB	8.7	9.1	12.2	16.9	18.7	22.4	21.7	21.6	19.6	17.5	13.9	10.4		
	2%	DB	9.2	11.9	18.6	24.0	27.7	31.4	32.8	33.2	28.2	22.7	15.9	9.8		
		MCWB	6.6	8.0	11.0	14.5	18.4	21.4	21.9	21.5	18.7	16.2	12.3	7.7		
	5%	DB	7.3	9.7	16.0	21.8	25.9	29.8	31.0	31.1	25.9	20.6	13.7	7.7		
		MCWB	5.1	6.8	9.7	13.5	17.6	20.7	21.4	20.8	17.9	15.0	10.9	6.1		
	10%	DB	5.6	7.7	13.6	19.8	23.7	27.9	29.3	28.9	23.7	18.2	11.8	6.4		
		MCWB	4.1	5.8	8.8	12.4	16.5	19.9	20.7	20.3	16.8	13.9	9.7	5.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.1	9.7	13.0	17.3	20.5	23.5	24.0	23.5	20.6	18.1	14.2	10.7		
		MCDB	11.8	13.7	19.2	25.5	27.3	31.1	31.1	30.9	28.7	24.0	17.6	12.7		
	2%	WB	7.0	8.2	11.6	15.3	19.2	22.2	22.9	22.4	19.5	16.7	12.7	7.9		
		MCDB	8.7	11.1	17.0	22.2	25.8	29.5	30.2	29.9	26.3	21.7	15.5	9.3		
	5%	WB	5.5	7.1	10.4	14.2	18.1	21.4	22.1	21.6	18.5	15.5	11.3	6.4		
		MCDB	6.9	9.3	15.1	20.3	24.3	28.3	29.3	29.1	24.4	19.5	13.4	7.5		
	10%	WB	4.1	6.0	9.1	13.1	17.1	20.5	21.2	20.9	17.5	14.4	10.0	5.2		
		MCDB	5.5	7.6	13.1	19.1	22.7	26.8	27.9	28.0	22.4	17.7	11.5	6.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.1	7.1	9.4	11.0	10.4	10.6	11.0	11.5	10.4	9.9	7.1	5.4		
		MCDBR	8.7	10.7	14.3	14.9	14.2	13.4	14.0	15.2	14.4	13.4	10.2	7.5		
	5% WB	MCWBR	6.5	7.3	8.1	7.2	6.3	5.5	5.5	5.7	6.3	7.4	6.8	5.8		
		MCWBR	8.5	9.3	12.8	13.5	12.7	12.3	12.5	13.4	12.7	12.3	9.6	7.1		
(40) Clear-Sky Solar Irradiance	taub		0.333	0.357	0.394	0.443	0.432	0.432	0.444	0.445	0.421	0.388	0.361	0.332		
		taud	2.360	2.306	2.230	2.110	2.200	2.234	2.202	2.212	2.242	2.315	2.344	2.370		
	Ebn at Noon		738	795	818	809	835	835	818	800	780	747	692	689		
		Edh at Noon	73	95	119	147	139	135	138	131	116	93	75	66		
(44) All-Sky Solar Radiation	RadAvg		1.06	1.88	3.11	4.58	5.53	6.00	5.90	5.29	3.72	2.41	1.29	0.81		
	RadStd		0.13	0.28	0.45	0.44	0.56	0.45	0.43	0.51	0.43	0.26	0.12	0.11		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (55 neighbors)		+0.62	N/A	N/A	N/A	+0.55	+0.52	-123	-180	+105	+57

Nomenclature: See separate page